

# Literature Status: Absolute Examples With No Uncountable Equilateral Set

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**Status.** `literature_already_answered`. This is a literature-status packet, not an original proof from the run.

**Source question.** Koszmider's 2015 paper *Uncountable equilateral sets in Banach spaces of the form  $C(K)$* , arXiv:1503.06356, constructs consistent nonseparable  $C(K)$  examples with no uncountable equilateral set. In the abstract and introduction it records the remaining absolute-example problem: whether there is a ZFC example of a nonseparable Banach space with no uncountable equilateral set. The same discussion also mentions sharper variants involving  $C(K)$  spaces and equivalent renormings.

**Supporting answer.** Koszmider's later paper *Banach spaces in which large subsets of spheres concentrate*, arXiv:2104.05335, proves in ZFC that there is a strictly convex Banach space of density continuum such that every uncountable subset of its unit sphere has an uncountable subset with all mutual distances below  $1 - \varepsilon$  for some  $\varepsilon > 0$ . The paper then invokes the standard Terenzi reduction from arbitrary equilateral sets to unit-sphere 1-equilateral sets and states the consequence as Theorem 2: there is a strictly convex Banach space of density continuum which does not admit an uncountable equilateral set.

**Identification.** This directly supplies the absolute nonseparable Banach-space example asked for in the 2015 discussion. The supporting paper explicitly compares this ZFC result with the earlier consistent  $C(K)$  examples from arXiv:1503.06356.

**Scope limitations.** This packet does not claim that the sharper absolute  $C(K)$  version is solved. It also does not answer the separate question whether there are consistent examples whose every equivalent renorming has no uncountable equilateral set.

## References

- [1] P. Koszmider, *Uncountable equilateral sets in Banach spaces of the form  $C(K)$* , arXiv:1503.06356.
- [2] P. Koszmider, *Banach spaces in which large subsets of spheres concentrate*, arXiv:2104.05335.